

# A Causal-Chain GraphRAG for Verifiable Provincial Flood Attribution in the Chao Phraya Basin

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**Abstract**—Flood question-answering systems are usually judged by predictive accuracy, yet a correct answer is not necessarily a verifiable one. This paper presents a GraphRAG system that answers the question of why a province flooded by traversing a hydrological causal chain (rainfall, reservoir, river reach, downstream province) in which every edge carries an explicit evidence property, so each prediction is traceable to a gauge, a dataset, and a timestamp. Prediction uses a de-circularized two-signal gate: a fluvial signal that fires when a reach exceeds its own two-year return stage, and a pluvial signal that fires when 3-day or 30-day rainfall exceeds the province's own ten-year return level. Ground truth is satellite flood extent, while the gate uses river gauges and reanalysis rainfall, so no threshold is derived from the labels. Across five Chao Phraya events (2021-2025; 23 provinces; 115 province-cases) the system attains a Matthews correlation coefficient (MCC) of +0.203 with traceability of about 0.90, and is the only system compared with skill above chance: an entity-graph baseline reaches a higher F1 (0.844 against 0.795) purely by predicting almost every province flooded, yet scores MCC 0.000 and specificity 0. A counterfactual analysis quantifies each mechanism, and agreement between mechanisms raises precision from 0.782 to 0.833. Three convergent tests show that the attainable MCC of about 0.20 is a limit of the data rather than of the method. We report where the method loses and disclose the correction of an earlier evaluation artifact that had inflated F1.

**Keywords**—graph retrieval-augmented generation, causal graph, flood attribution, faithfulness, Matthews correlation coefficient, de-circularized evaluation, Chao Phraya basin

## I. INTRODUCTION

Flood question-answering and early-warning systems are commonly evaluated with accuracy-oriented scores such as F1 or the critical success index. For operational decision support, however, the verifiability of an explanation matters as much as its correctness: an operator asked why a particular province is expected to flood must be able to follow the answer back to concrete measurements. Recent work on retrieval-augmented generation makes the same distinction, showing that a system can be correct for reasons it cannot expose, so that correctness and faithfulness are separate properties [3].

This paper addresses that gap for provincial flood attribution in the Chao Phraya basin of Thailand. We build a causal-chain GraphRAG in which the graph encodes the physical route water takes, and in which every edge stores the evidence that justifies it, so a prediction can be replayed as a chain of measurements rather than presented as an opaque score. Fig. 1 summarizes the system. The main contributions of this paper are: (1) a causal-chain GraphRAG for flood attribution with complete edge-level evidence traceability; (2) a de-circularized prediction gate whose thresholds are drawn from gauge and rainfall climatology rather than from the

labels, together with evidence that at high flood base rates F1 is gamed by a trivial predict-all baseline, so that MCC, specificity, and traceability are the appropriate criteria; (3) a counterfactual necessity and sufficiency analysis that quantifies how much each hydrological mechanism contributes; and (4) three convergent tests showing that the attainable skill is bounded by the data rather than by the combiner, together with a transparent account of correcting an earlier over-flagging artifact.

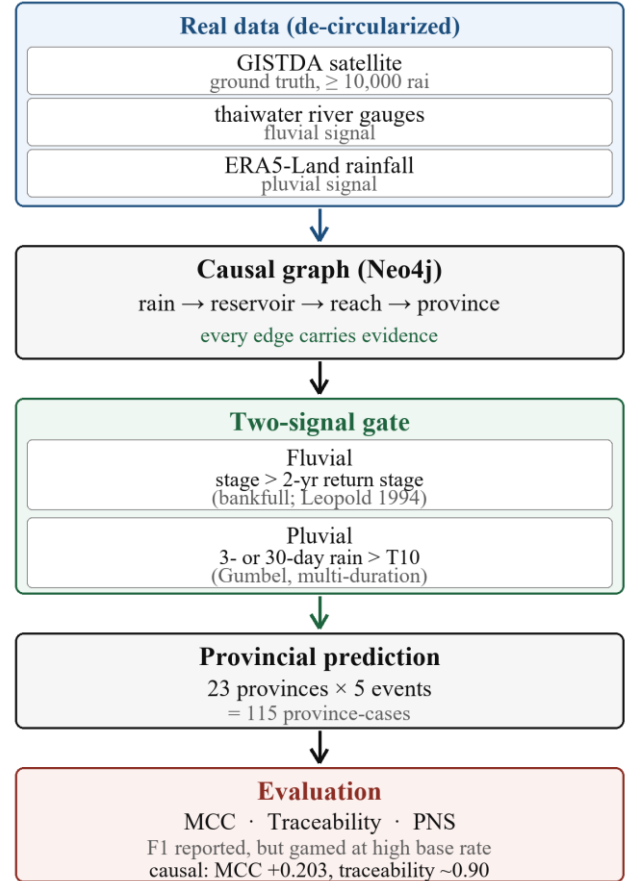


Fig. 1. System overview: de-circularized data sources, the causal graph whose edges carry evidence, the two-signal gate, and the evaluation layer.

## II. RELATED WORK

### A. GraphRAG and Faithfulness in Retrieval

Graph-based retrieval-augmented generation organizes retrieval over an explicit graph so that multi-hop questions can be answered by traversal rather than by similarity alone [1], and systematic comparisons report that the advantage of graph traversal widens as the number of required hops grows [2].

Evaluation of such systems increasingly separates whether an answer is correct from whether it is attributable to the retrieved evidence [3]. Our system takes the strict form of this idea: the graph is not only the retrieval substrate but also the explanation, because every edge on a returned path carries the source of the fact it asserts.

#### B. Causal and Network Models of River Basins

A river basin is naturally represented as a directed graph whose edges follow the direction of flow, and causal structure has been inferred for extreme discharges over such a basin network [9]. Compared with that line of work, our graph is deliberately small and physical: its nodes are rain stations, reservoirs, river reaches, confluences, and provinces, and its prediction rule is a fixed hydrological gate rather than a learned scorer, which is what allows the evaluation to remain de-circularized.

#### C. Evaluation under Class Imbalance and Counterfactual Attribution

For binary problems with skewed class balance, the Matthews correlation coefficient is preferred to F1 and accuracy because it accounts for all four cells of the confusion matrix and penalizes a classifier that never rejects the majority class [4]. Separately, counterfactual accounts of explanation formalize how necessary and how sufficient a factor is for an outcome [5]; we adopt that framing to measure the contribution of each hydrological mechanism by intervening on the gate.

#### D. Hydrological Thresholds

The thresholds in our gate are taken from established hydrology rather than fitted. Bankfull stage, at which water begins to spread onto the floodplain, recurs on average about every one and a half to two years [6], which motivates a two-year return stage as the fluvial trigger. Return-period rainfall is estimated with Gumbel frequency analysis, and the 2011 Chao Phraya flood has been assigned a return period of roughly ten to twenty years [7], which motivates a ten-year return level as the pluvial trigger. Finally, probability calibration is known to be unreliable when few samples are available [8], which is why our calibration results are reported as inconclusive rather than as a positive finding.

### III. PROPOSED SYSTEM

#### A. System Overview

As shown in Fig. 1, the system has three parts: a causal graph that stores the basin and the evidence behind each link, a two-signal gate that turns measurements into a provincial prediction, and an evaluation layer that scores the prediction against satellite ground truth. A province is predicted to flood when either signal fires and the province is not protected by dikes, as expressed in (1). No structural parameter is learned; the only free choices are the two return periods, and both are fixed a priori from the hydrological literature.

#### B. Causal Graph

The graph contains five node types, namely RainStation, Reservoir, RiverReach, Confluence, and Province, connected by directed edges that follow the direction of water flow: FEEDS, RUNOFF\_TO, OVERFLOWS\_TO, FLOWS\_TO, and INUNDATES. Every edge stores an evidence property recording the station, dataset, and timestamp that justify it, and a guard in the loader rejects any edge without one, so traceability is a structural invariant rather than a reporting

convention. Hop count is the length of the shortest causal path from a rainfall source to a province, and is a property of the basin geometry, so the hop buckets used in evaluation are fixed across events. The graph covers eight Chao Phraya sub-basins and 23 provinces.

#### C. De-circularized Two-Signal Gate

The fluvial signal snaps each river reach to the gauges physically located on it and declares the reach over-topped when its 95th-percentile stage during the flood season exceeds that gauge's own two-year return stage, estimated as the median annual maximum computed leave-one-event-out. The 95th percentile is used instead of the maximum because isolated telemetry spikes would otherwise trigger the gate. The main-channel bank recorded for these gauges sits high above the floodplain and is rarely reached, so using it directly would make the gate almost silent; the two-year return stage is the standard hydrological proxy for the onset of overbank flow [6]. The pluvial signal fires when a province's peak 3-day or 30-day accumulated rainfall exceeds its own ten-year Gumbel return level, the shorter duration capturing cloudbursts and the longer one capturing prolonged rain that saturates the basin. Crucially, ground truth is satellite flood extent whereas the gate reads river gauges and reanalysis rainfall, and every threshold is derived from the climatology of the signal itself, so no label information enters the decision rule.

$$\text{flood}(p) = [ S \geq S_2 \vee \max_n R_n \geq R_{n,10} ] \wedge \neg \text{protected} \quad (1)$$

In (1),  $S$  is the 95th-percentile stage of the gauge governing the reach that inundates province  $p$  and  $S_2$  is that gauge's two-year return stage;  $R_n$  is the peak  $n$ -day rainfall over  $p$  for  $n$  in  $\{3, 30\}$  and  $R_{n,10}$  the corresponding ten-year return level; and  $\text{protected}$  marks provinces shielded by dikes. The rule has no tunable weight.

#### D. Baselines

Two baselines share the province universe and the evaluation set. The entity-graph baseline retrieves over an entity-relation graph without the causal gate, and the vector baseline ranks a corpus of Thai flood news with character  $n$ -gram TF-IDF, a tokenization chosen because Thai text has no delimited word boundaries. Neither baseline can attach evidence to a prediction, so their traceability is zero by construction.

### IV. EXPERIMENTAL SETTING

The evaluation covers five Chao Phraya flood events from 2021 to 2025 over the same 23 provinces, giving 115 province-cases. Ground truth is the per-province flooded area derived from GISTDA satellite products [10], with a threshold of 10,000 rai fixed before any experiment; on this definition the base rate of flooding is about 0.73, which is what makes the choice of metric consequential. The fluvial signal is computed from public thaiwater river-gauge time series [11], the pluvial signal from ERA5-Land daily precipitation obtained through the Open-Meteo archive [12], provincial geometry from GADM 4.1 [13], and an independent check of flow direction from the Copernicus GLO-90 digital elevation model [14].

We report precision, recall, specificity, F1, and MCC, and treat MCC as primary for the reason given in Section II.C. Traceability is the fraction of the claims in an explanation that resolve to an evidence property on the traversed edges. For each mechanism we compute a probability of necessity,

defined as the fraction of true positives lost when that mechanism is removed by intervention, and a probability of sufficiency, defined as its precision when it fires; we also report precision as a function of how many mechanisms agree. Return levels are estimated leave-one-event-out so that an event never contributes to its own threshold. The pipeline contains no learned structural parameters and no random component, so repeated runs reproduce identical outputs given the same data snapshot.

## V. RESULTS AND DISCUSSION

### A. Main Comparison

TABLE I. PROVINCIAL FLOOD ATTRIBUTION, POOLED OVER FIVE EVENTS (N = 115)

| System          | F1    | MCC    | Spec. | Rec. | Trace. |
|-----------------|-------|--------|-------|------|--------|
| causal-graphrag | 0.795 | +0.203 | 0.387 | 0.81 | ~0.90  |
| entity-graphrag | 0.844 | 0.000  | 0.000 | 1.00 | 0      |
| vector-rag      | 0.096 | -0.497 | 0.516 | 0.06 | 0      |

Table I reports the pooled comparison. The entity baseline obtains the highest F1, but it does so by predicting that almost every province floods: its recall is 1.00, its specificity is 0.000, and its MCC is exactly 0.000, which is the value expected from a classifier with no discriminative power. The causal system is the only one of the three with MCC above zero and the only one whose predictions carry evidence. Per-event MCC is +0.47, +0.52, +0.04, +0.37, and +0.17 for 2021 through 2025; we report all five values, including the weak 2023 result, in which the gauges and rainfall of that year gave little signal.

### B. Mechanism Necessity and Sufficiency

TABLE II. MECHANISM NECESSITY AND SUFFICIENCY UNDER INTERVENTION

| Mechanism            | Necessity | Suffic. | $\Delta$ Recall |
|----------------------|-----------|---------|-----------------|
| Fluvial (2-yr stage) | 0.35      | 0.79    | -0.29           |
| Pluvial 30-day       | 0.21      | 0.77    | -0.17           |
| Pluvial 3-day        | 0.00      | 0.85    | 0.00            |

Table II decomposes the gate by intervention. The fluvial signal is the most necessary: removing it costs 0.29 of recall. The 30-day pluvial term is the second most necessary and is what recovers prolonged-rain events such as 2023. The 3-day term has necessity 0.00, meaning every province it catches is already caught by another mechanism, yet its sufficiency of 0.85 is the highest of the three, so it is precise but redundant on this data. Precision also increases with agreement: 0.782 when at least one mechanism fires, 0.833 when at least two agree, and 0.667 for the small set where all three agree. Agreement between independent physical mechanisms therefore provides a confidence signal that is itself traceable, which neither baseline can offer.

### C. A Ceiling in the Data, Not the Method

TABLE III. PREDICTABILITY-CEILING TESTS (POOLED MCC)

| Test                                           | MCC          |
|------------------------------------------------|--------------|
| Learned logistic combiner (in-sample; LOEO)    | 0.000; 0.000 |
| Four principled rule variants                  | 0.13 to 0.19 |
| Ground-truth threshold sweep (10k to 200k rai) | 0.21 to 0.14 |
| Proposed gate (fluvial and pluvial)            | +0.203       |

A natural question is whether the modest MCC reflects a weak combiner or a weak signal. Table III reports three tests that point to the latter. First, a logistic model trained on the continuous signal ratios degenerates to predicting every

province, scoring MCC 0.000 both in-sample and leave-one-event-out, so no learned combination of these features beats the fixed rule at this base rate. Second, four principled rule variants cluster between 0.13 and 0.19, and dropping the redundant 3-day term changes nothing, which is consistent with the necessity analysis. Third, raising the ground-truth threshold from 10,000 to 200,000 rai lowers the base rate but does not raise MCC, so the limitation is not imbalance alone: these signals do not separate flood severity. We therefore read the attainable MCC of about 0.20 as a property of the available measurements rather than of the model family.

### D. Early-Warning Extension

As an extension we also treated the gate as an early-warning rule. Its binary skill is a probability of detection of 0.81, a false-alarm ratio of 0.218, and a critical success index of 0.660. Calibrating a probability by causal-hop depth yields a Brier skill score over climatology of about +0.04, but with five events the event-level cluster bootstrap gives a 95% confidence interval from -0.54 to 0.16 and  $p = 0.26$ . We therefore report the calibration as inconclusive; it is consistent with the known difficulty of calibrating probabilities from few samples [8].

### E. Discussion and Limitations

Taken together, the results support the two hypotheses in a specific and limited sense. H1 holds: traceability is 0.90 for the causal system and zero for both baselines, and this gap is structural rather than incidental, because neither baseline stores evidence on the objects it retrieves. H2 holds in that the prediction is a basin-wide footprint derived from event state, so its quality does not decay as the causal chain lengthens. The most consequential measurement insight is that F1 is actively misleading here: at a base rate of 0.73 a predict-all rule earns F1 0.844 while contributing no information, and only MCC and specificity expose that. A second insight concerns what the system still misses. The false negatives concentrate in mainstem provinces in 2023 and 2024, and a soil-moisture probe using ERA5-Land root-zone data shows that those provinces were not saturated, with a median event percentile of 28% in 2023 and 38% in 2024 and Nakhon Sawan between the 0th and 7th percentile. Their flooding is therefore not of the saturation-driven type the pluvial signal is designed to detect. A backwater effect at confluences, or inundation of floodplain below the main-channel bank, would be consistent with this pattern, but we did not run a hydrodynamic model and so we state it as a hypothesis rather than a finding.

Four limitations should be noted. First, the absolute skill is modest: an MCC of about 0.20 means the system is informative but far from reliable at province level, and the operating point reported here favours recall, so specificity is 0.387. Second, five events give little statistical power; the early-warning calibration in Section V.D is not significant, and per-event MCC varies from +0.04 to +0.52. Third, the ground truth is a thresholded satellite product, so a province just below 10,000 rai is scored as dry although it may have flooded locally. Fourth, and most important for interpretation, an earlier version of this evaluation used a coarser gate that marked an entire sub-basin as flooding whenever any gauge in it over-topped. That gate reported an F1 near 0.9, but it did not reproduce under the per-reach gate and contradicted the physical record, since the mainstem gauge at Nakhon Sawan never exceeded its bank between 2021 and 2025. Replacing it lowered F1 to 0.548, and the threshold corrections described in Section III.C raised it to 0.795. We report the earlier and the

corrected numbers together rather than silently replacing them.

## VI. CONCLUSION

This paper presented a causal-chain GraphRAG for provincial flood attribution whose predictions are traceable to the measurements that justify them, and whose decision rule takes no threshold from the labels it is scored against. On five Chao Phraya events the system is the only one of three compared that combines complete traceability with skill above chance, at an MCC of +0.203, while an entity-graph baseline achieves a higher F1 with no skill at all. A counterfactual analysis showed which hydrological mechanism carries the prediction and that agreement between mechanisms raises precision, and three convergent tests indicated that the attainable skill is bounded by the available measurements rather than by the combiner. The most useful direction for future work therefore concerns data rather than modelling: a hydrodynamic treatment of confluence backwater, denser gauging where the mainstem misses are concentrated, evaluation at district rather than province resolution to reduce the base rate, and additional events to give the calibration enough power to be decided.

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